FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. Least count $= \frac{0.5}{100} = 0.005\,mm$

Diameter will be $0.5\times 6 + 0.005\times 74 - 0.5\times 0.005 = 3.345\,mm$

2. A

Sol. $$\frac{4\lambda D}{d} = \frac{3\lambda\left(D + \frac{1}{2}a_0 t^2\right)}{d}$$

$$t = \sqrt{\frac{2D}{a_0}}$$

3. D

Sol. Direction of electric field is along direction of current

$$\frac{2\sigma}{\tan\theta} = \frac{\sigma}{\tan 30^\circ}$$

$$\theta = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$$

4. B

Sol. $$\frac{Q}{4\pi \in_0 KR^2} = E_0 = 8$$

$$\Delta V = \frac{Q}{4\pi \in_0 K}\frac{(2R-R)}{R\times 2R}$$

$$= \frac{Q}{8\pi \in_0 KR} = 2Volt$$

5. C

Sol. Maximum possible loss occurs in case of perfectly inelastic collision loss in

$$KE = \frac{1}{2}\mu(V_r)^2 - \frac{1}{2}\mu(U_r)^2$$

$$= \frac{1}{2}\times\frac{3}{4}\times\left(\frac{8}{3}\right)^2 - \frac{1}{2}\times\frac{3}{4}\times\left(\frac{4}{3}\right)^2 = 2\,Joule$$

6. C

Sol. $B2\pi r = \mu_0 I_{in}$

$$Kr^2 2\pi r = \mu_0\int_0^r J2\pi r dr$$

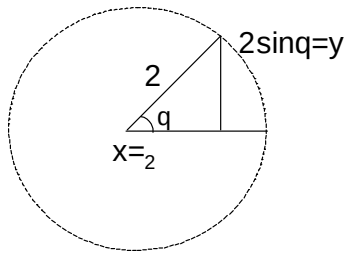
On differentiating both side

$$K\frac{d(r^3)}{dr} = \mu_0 Jr, \frac{dj}{dr} = \frac{3K}{\mu_0}$$

7. B

Sol. Since particle moves in xy plane

So, z = 0



$$d\vec{r} = dx\hat{i} + dy\,\hat{j}$$

$$dw = \vec{f}.\,d\vec{r} = \int (x^2 - y)dx + (x+5y)dy$$

$$w = \int dw = 4\int_0^{2\pi} d\theta = 8\pi$$

8. AC

Sol. Let I_0 be minimum current required to flip, then

$$I_0\ell^2 B = \frac{4mg\ell}{2}$$

$$I_0 = \frac{2mg}{\ell B}$$

$$I = \frac{4mg}{\ell B} = 2I_0$$

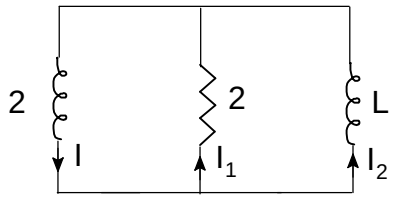
Net torque about one edge

$$\frac{5m\ell^2}{3}\alpha = I\ell^2 B - \frac{4mg\ell}{2} \quad \Rightarrow \alpha = \frac{6g}{5\ell}$$

$$a_{cm} = \frac{\alpha\ell}{2} = \frac{3g}{5}, \quad N - 4mg = \frac{4m3g}{5}, \quad N = \frac{32mg}{5}$$

9. BCD

Sol. After time 't'



$$2L\frac{dI}{dt}+L\frac{dI_2}{dt}=0$$

$$2\int_{E/R}^{I} dI+\int_{0}^{I} dI_2=0, \qquad I=\frac{2E}{3R}$$

$$2L\frac{dI}{dt}+IR=0$$

$$2L\frac{dI}{dt}+\frac{dq}{dt}2R=0$$

$$2L\int_{E/R}^{2E/3R} dI+\int_{0}^{Q} dq\,2R=0$$

$$Q=\frac{LE}{3R^2}$$

$$\Delta H=U_i-U_f=\frac{1}{2}2L\left(\frac{E}{R}\right)^2-\frac{1}{2}3L\times\frac{4E^2}{9R^2}=\frac{LE^2}{3R^2}$$

$$U_L=\frac{1}{2}L\left(\frac{2E}{3R}\right)^2=\frac{2LE^2}{9R^2}$$

10. BD

Sol. Expansion of body is like photographic enlargement.

11. ACD

Sol. Angular momentum of rigid body about any point is given as

$$L=\overrightarrow{r_{cm}}\times m\vec{V}_{cm,p}+I_{cm}\vec{\omega}$$

For Ring I

$$L_P=\left(mR^2+mR^2\right)\omega=2mR^2\omega$$

$$L_Q=2m\omega R\times 4R+mR^2\omega=9mR^2\omega$$

For Ring II

$$L_P=4m\omega R\times 4R-mR^2\omega=15m\omega R^2$$

$$L_Q=\left(mR^2+mR^2\right)\omega=2mR^2\omega$$

12. AD

Sol. Image will be at rest with respect to object when velocity of particle is parallel to mirror

$$\tan 30^\circ=\frac{10\sqrt{3}-10t}{10}\Rightarrow t=\frac{2}{\sqrt{3}}\,Second$$

13. ABD

Sol. $E\propto\frac{1}{n^2},\ r\propto n^2,\ P\propto\frac{1}{n}$

14. AC

Sol. Let angular velocity of disc A and B be ω_1 and ω_2 respectively. Component of velocity of points on rod along rod would be same

$\omega_2 R\cos 30^0 = 4\omega_0 R\cos 30^0$

$\omega_2 = 4\omega_0$

$\omega_1 R\cos 30^0 = \omega_2 R\cos 60^0$

$\omega_1 = \dfrac{4\omega_0}{\sqrt{3}}$

$K = \dfrac{1}{2}\times\dfrac{3}{2}mR^2\omega_1^2 + \dfrac{1}{2}\times\dfrac{3}{2}mR^2\omega_2^2 + \dfrac{1}{2}\times\dfrac{3}{2}m16R^2\omega_0^2$

$= 28m\omega_0^2R^2$

15. B

Sol. Electric potential on surface of conductor = $\dfrac{KQ}{2a}+\dfrac{KQ}{4a}=\dfrac{3KQ}{4a}$

Electric potential at centre = $\dfrac{2KQ}{a}-\dfrac{KQ}{a}+\dfrac{3KQ}{4a}=\dfrac{7KQ}{4a}$

16. A

Sol. When conductor is earthed potential becomes zero. Let charge on outer surface of conductor be q. $\dfrac{Kq}{2a}+\dfrac{KQ}{4a}=0,\ q=-\dfrac{Q}{2}$

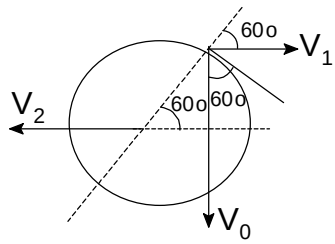
So, amount of charge flowing through wire $= Q-\left(-\dfrac{Q}{2}\right)=\dfrac{3Q}{2}$

17. A

Sol. Velocity acquired by particle will be $V_0=\sqrt{2g\times 20}=20\,m/s$

$V_0\cos 60^\circ = V_2\sin 60^\circ$

$V_1=\dfrac{20}{\sqrt{3}}$



From conservation of momentum for system of particle along horizontal

$1V_1 = 2V_2 \quad \Rightarrow V_2=\dfrac{10}{\sqrt{3}}, \quad e=\dfrac{\frac{V_1}{2}+\frac{V_2}{2}}{\frac{V_0\sqrt{3}}{2}}=\dfrac{1}{2}$

18. B

Sol. Let N be normal force exerted by inclined surface on sphere

$\int N\cos 30^\circ = 2\times 2V_2$

$\int Ndt=\dfrac{80}{3}$

Chemistry

PART – II

SECTION – A

19. A

Sol. C will formed at will the rate of B

K of A is = $K_1 + K_2 = \frac{0.693}{3}$

Let at time t [C] = 2[A] and [B] = X %

So, $2x = (100 - 3x) \times 2$

$\Rightarrow x = 25$

applying 1st order kinetics

$\frac{\ln_2}{3} \times t = \ln \frac{100}{100 - 3x} \Rightarrow t = 6n$

20. B

Sol. $0.209 = 0.80 + \frac{0.0591}{1} 10 g KSP$

$KSP = 10^{-10}$

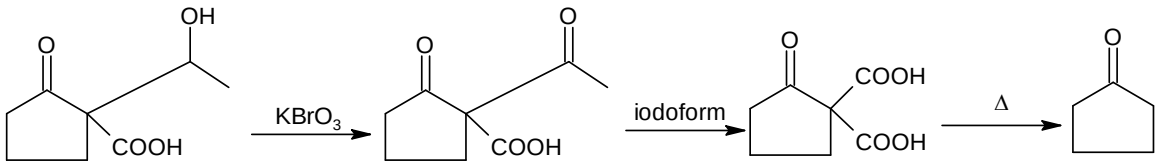
$\Rightarrow$ solubility of AgCl in 0.01M solution is x

KSP = x(x + 0.1)

$\Rightarrow x = 10^{-8}$, mole of AgCl dissolve in 10L = 10^{-7} z = 7

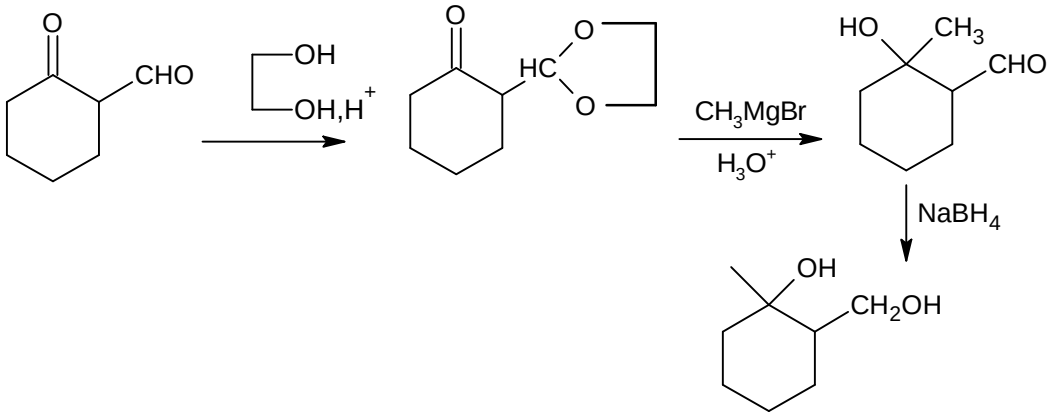
21. C

Sol.



22. A

Sol.



23. B

Sol. For X apply buffer formula of ph, we get [salt] = $5 \times 10^{-2} M$ for y $[salt]_2$ = 5M on mixing equal volume, final [Acid] = 0.5 M & [sat] = 2.5025 so $pH = 5 + \log \frac{2.5025}{0.5} = 5.7$

24. D

Sol. $M \rightarrow M^{+} + e^{-}$

$n = \frac{5}{25} = 0.2$ energy required to form M^{+}ions = 0.2 × 800 = 160 kj/mole

Remaining 90 kj used in the conversion of M^{+} to M^{+2}

So no of mole of M^{+2} formed = $\frac{90}{1500} = 0.06$

$\%M^{2+} = \frac{0.06}{0.2} \times 100 = 30\%$

25. B

Sol. $P_1V_1 = P_2V_2$, so expanded vel^{m} = 80L
available volume to fill the ballons = 80 – 8 =72 L
so no of ballons = 18

26. ABC

Sol. fact

27. ABC

Sol. conceptual

28. CD

Sol. Total line for H atom and li^{2+} ion are 6,15. But 2 → 1 has same energy as 3 → 6 in li^{2+}ions so total line = 20 if sample is containing only H then there are 2 balmer lines in li^{2+} all lyman lines are in u.v = 5 all balmer lines in u.v = 4 in li^{2+} few more lines can be in u.v

29. ABC

Sol. $Pt^{+2}, Pd^{+2} Au^{+3}$ for I.O.C dsp^2 with SFL and WFL when CN = 4 $Hg^{2+}(Sd^{10})$ always form outer orbital complex Sp^3 when CN = 4

30. BC

Sol. manganese, Tungsten

31. ABD

Sol. fact

32. AC

Sol. $N(Me)_3$ is pyramidal while $Ni(Sime_3)_3$ is trigonal planar

33. C

Sol. Only Fe^{2+} will oxidice, so $25 \times [Fe]^{2+} = 25 \times 0.02 \times 5$ M = 0.10 M

Total Fe^{2+} in second part $25 \times y = 40 \times 0.02 \times 5$ y = 0.16 M = 0.16N

So, Fe^{3+} = 0.16 – 0.10 $\Rightarrow$ 0.06M

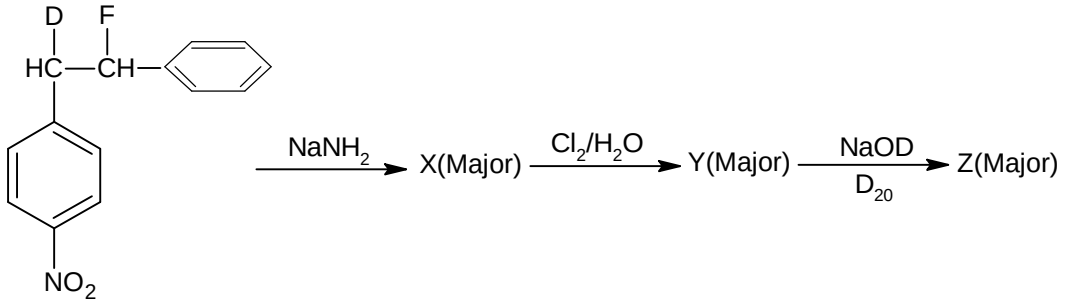
34. A

Sol. Only Fe^{2+} will oxidice, so $25 \times [Fe]^{2+} = 25 \times 0.02 \times 5$ M = 0.10 M

Total Fe^{2+} in second part $25 \times y = 40 \times 0.02 \times 5$ y = 0.16 M = 0.16N

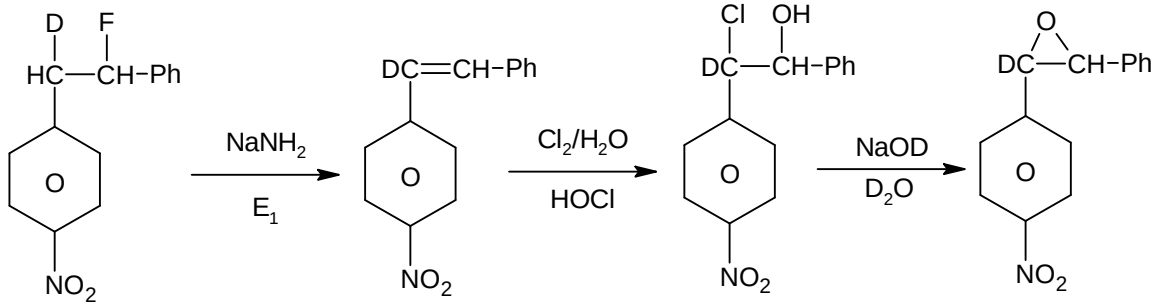
So, Fe^{3+} = 0.16 – 0.10 $\Rightarrow$ 0.06M

P – 11



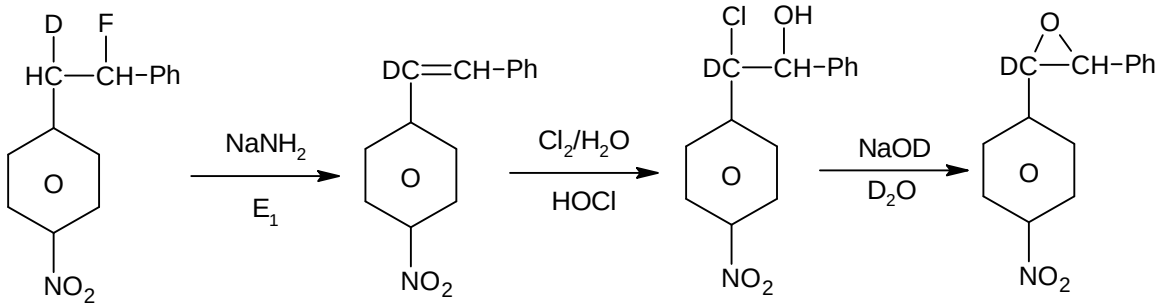
35. D

Sol.



36. B

Sol.



Mathematics

PART – III

SECTION – A

37. B

Sol. $z\omega = |z|^2 \Rightarrow \omega = \overline{z}$

$|z + \overline{z}| + |\omega + \overline{\omega}| = 4$

$|x| + |y| = 2$

which is a square $\therefore$ Area = 8 sq. units

38. D

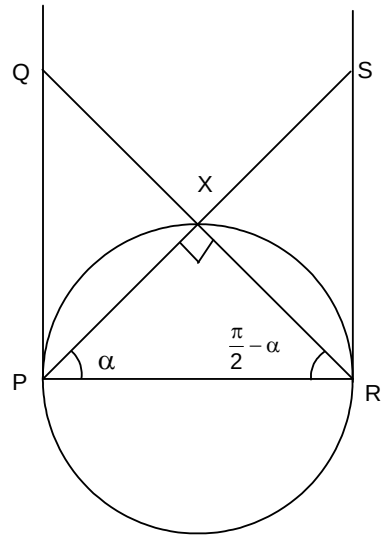
Sol. Let $S = 1 + 5\rho + 9\rho^2 + ... + (4n-3)\rho^{n-1}$

Then $\rho S = \rho + 5\rho^2 + ... + (4n-7)\rho^{n-1} + (4n-3)\rho^n$

Subtracting

$(1-\rho)S = 1 + 4\rho + 4\rho^2 + ... + 4\rho^{n-1} - (4n-3)\rho^n$

Hence $S = \dfrac{4n}{\rho - 1}$ Taking $\rho = e^{2\pi i/n}$



We have

$$1 + 5e^{2\pi i/n} + 9e^{4\pi i/n} + ... + (4n-3)e^{2(n-1)\pi in} = \frac{4n}{e^{2\pi i/n} - 1}$$

$$\frac{4n\left\{\left(\cos\frac{2\pi}{n}\right) - 1 - i\sin\frac{2\pi}{n}\right\}}{\left(\cos\frac{2\pi}{n} - 1\right)^2 + \sin^2\left(\frac{2\pi}{n}\right)} = \frac{n\left(\cos\frac{2\pi}{n} - 1 - i\sin\frac{2\pi}{n}\right)}{\sin^2\frac{\pi}{n}}$$

We see that

$$1 + 5\cos\left(\frac{2\pi}{n}\right) + 9\cos\left(\frac{4\pi}{n}\right) + ... + (4n-3)\cos\left(\frac{2(n-1)\pi}{n}\right) = -2n$$

$$5\sin\frac{2\pi}{n} + 9\sin\left(\frac{4\pi}{n}\right) + ... + (4n-3)\sin\left(\frac{2(n-1)\pi}{n}\right) = -2n\cot\left(\frac{\pi}{n}\right)$$

$\therefore$ a – b – c = 4 + 2 + 2 = 8.

39. C

Sol. $$I_1 = 2\int_1^{10^2} \{t\}dt \quad \left[\text{Put } \sqrt{x} = t,\ \frac{dx}{2\sqrt{x}} = dt\right]$$

$$I_2 = \int_1^{10} \left(x\{x^2\}\right)dx \qquad [x^2 = N,\ 2x\,dx = dN]$$

$$I_2 = \int_0^{10^2} \frac{\{N\}dN}{2}$$

$$\frac{I_1}{I_2} = 4, \quad I_1 = 200\int_0^1 \{t\}dt$$

$$I_1 = 200\left[\frac{x^2}{2}\right]_0^1 = 100$$

40. A

Sol. $f'(x) = (a-1)(x^2+x+1)^2 - (a+1)(x^4+x^2+1)$

for $f'(x) = 0$

$(x^2+x+1)[(a-1)(x^2+x+1) - (a+1)(x^2-x+1)] = 0$

$\Rightarrow$ either $x^2+x+1=0$ or $x^2-ax+1=0$

For distinct roots, $D > 0$.

41. D

Sol. $f(x)$ is not differentiable at $x = \frac{\pi}{4}$

$\therefore$ option (D) is incorrect.

42. C

Sol. Clearly $X = \{100, 200, 300, 400, 500, 600, 700, 800, 900\}$

And for the equation $x^3 - 3ax^2 + (298a + 299)x - 2 = 0$ to have positive point of maxima. The equation $f'(x) = 0$ should have both the roots positive and unequal, which gives $a > 299$.

$\therefore$ $Y = \{a; a > 299\}$

$\therefore$ Number of elements in $X \cap Y = 7$.

43. D

Sol. $|\cos x| + \cos x > 0 \rightarrow x \in \left[0, \frac{\pi}{2}\right)$

For $x \in \left[0, \frac{\pi}{2}\right)$

$1 \le \sin x + \cos x \le \sqrt{2}$

But $\sin x + \cos x \neq 1 \Rightarrow x \in \left(0, \frac{\pi}{2}\right)$

Now for $x \in \left(0, \frac{\pi}{2}\right)$

$\log_{\sin x + \cos x}(|\cos x| + \cos x) \ge 0$

$\Rightarrow \quad \cos x \ge \frac{1}{2} \Rightarrow x \in \left(0, \frac{\pi}{3}\right]$

44. AD

Sol. Vertex of $f(x)$ will be at $x = \frac{a}{2}$

Case – I : If $a < 0$, then maximum value of $f(x)$ is at $x = 0$

$\therefore$ $f(0) = -5 \Rightarrow a^2 + 2a - 3 = 0$

$\Rightarrow$ $a = -3$ or $1 \quad \Rightarrow a = -3$

Case – II : If $a > 2$, then maximum value of $f(x)$ is at $x = 1$

$\therefore$ $f(1) = -5 \Rightarrow a^2 + a - 2 = 0$

$\Rightarrow$ $a = -2$ or 1 which is not possible.

Case – III : If $0 < a < 2$, then maximum value is at $x = \frac{a}{2}$

$\Rightarrow f\left(\frac{a}{2}\right) = -5 \quad \Rightarrow 3a^2 + 8a - 12 = 0 \quad \Rightarrow a = \frac{-4 \pm 2\sqrt{13}}{3}$

$\Rightarrow a = \frac{-4 + 2\sqrt{13}}{3}$ is possible

45. ACD

Sol. $Y - y - \frac{dy}{dx}(X - x)$

$$A\left(x - \frac{y}{\frac{dy}{dx}}, 0 \right), \quad B(x, 0)$$

Q — B — A

$$x - \frac{y}{\frac{dy}{dx}} = 2x, \quad -y = x\frac{dy}{dx}$$

$$\frac{dy}{y} = -\frac{dx}{x}$$

ln |y| = –ln|x| + c

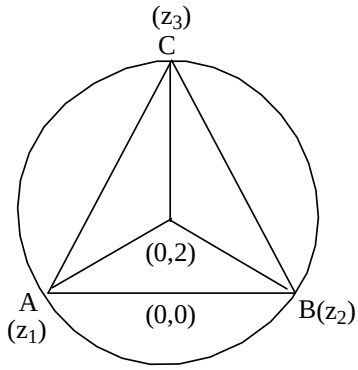
curve is passing from (1, 1), c = 0

xy = 1

46. AC

Sol. $\frac{z_2 - z_0}{z_1 - z_0} = e^{i2\pi/3}$

$$\frac{z_3 - z_0}{z_1 - z_0} = e^{-i\frac{2\pi}{3}}$$



47. AC

Sol. $x^2 + 3x + 2 > 0$

$\Rightarrow \quad x \in (-\infty, -2) \cup (-1, \infty)$

$x^2 + 3x + 2 \le 1$

$\Rightarrow \quad x \in \left[\frac{-3-\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2} \right]$

$x \in \left[\frac{-3-\sqrt{5}}{2}, -2 \right) \cup \left(-1, \frac{-3+\sqrt{5}}{2} \right]$

$0 < \sin^{-1}\left(\sqrt{x^2 + 3x + 2} \right) \le \frac{\pi}{2}$

$-\infty < \log\left(\sin^{-1}\left(\sqrt{x^2 + 3x + 2} \right) \right) \le \log\frac{\pi}{2}$

$\therefore$ set of all non positive integers.

48. ABCD

Sol. $\sin 3x = \cos x = \sin x\left(\frac{\pi}{2} - x\right)$

$3x = n\pi + (-1)^n\left(\frac{\pi}{2} - x\right)$

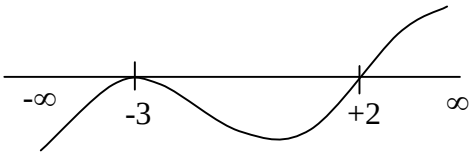
If n is even $x = \frac{m\pi}{2} + \frac{\pi}{8}$

If n is odd $x = m\pi + \frac{\pi}{4}$

$-\frac{\pi}{2} \le x \le \frac{\pi}{2}, \quad x = \frac{\pi}{8}, \frac{\pi}{4}, \frac{-3\pi}{8}$

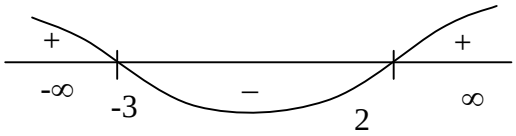
49. ABD

Sol. If n is even



$x \in (-\infty, -3) \cup (-3, +2)$

If n is odd



$x \in (-3, +2)$

Options (A), (B), (D) are correct.

50. AC

Sol. $|x^2 - 1| \ge 0$

$\Rightarrow a - 1 \ge 0 \Rightarrow a \ge 1$

$a \in [1, \infty)$

Again, $|x^2 - 1| = a - 1$

$\Rightarrow x^2 - 1 = \pm(a - 1)$

$\Rightarrow x^2 = 1 \pm (a - 1)$

$x^2 = 1 + a - 1 = a$ and $1 - a + 1 = 2 - a$

$a \ge 0$ and $2 - a \ge 0 \Rightarrow a \le 2$ (since $x^2 \ge 0$)

$\Rightarrow 0 \le a \le 2$

$a \in [0, 2]$.

51. A

Sol. Equation of tangent at $P\left(2t, \frac{2}{t}\right)$ is

$x + yt^2 = 4t$.

A = (4t, 0)

$B = \left(0, \frac{4}{t}\right)$

Hence area = 16 sq. units (constant).

52. C

Sol. s xy = 4, which is a rectangular hyperbola.

Hence $e = \sqrt{2}$.

53. C

Sol. Let $f(t) = 2t^3 - 9t^2 + 30 - a$

$f'(t) = 6t^2 - 18t = 0 \Rightarrow$ t = 0, 3

$\therefore$ For three real and distinct roots

f(0). F(3) < 0

(30 – a) (3 – a) < 0

$\Rightarrow$ a $\in$ (3, 30)

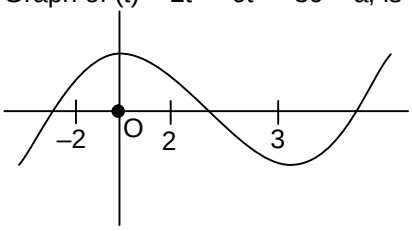
54. B

Sol. $x + \frac{1}{x} = t$ gives six real and distinct values then

Equation (1) should have three real and distinct roots and equation (2) should have two real distinct roots

3 < a < 30 …… (1)

Graph of $(t) = 2t^3 - 9t^2 + 30 - a$, is



From the graph it is clear that

f(- 2) > 0 and f(2) > 0

$\Rightarrow$ a < - 22 and a < 10

$\Rightarrow$ a $\in$ (- ∞, - 22) …… (2)

From (1) and (2) a $\in \phi$.